

Aleksandar Slavov

Software tools & AI safety research

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OpenReview: Aleksandar Slavov | Bulgaria

Work spanning Python tools for machine-learning debugging and memory analysis, and workshop research on instruction authority and prompt-injection defenses. Public code, examples, and paper links are provided below.

SELECTED SOFTWARE

ReproReduce

Python / PyTorch | [Repository](#)

Reduces failing Python and PyTorch programs to smaller standalone reproducers while preserving a configured failure. Candidate programs run in isolated subprocesses, with repeated validation and normalized failure fingerprints. Exports include the reproducer and environment metadata.

Public source, examples, tests, and benchmark reports.

jaxoom

Python / JAX | Experimental | [Repository](#)

Static and compiler-backed memory analysis for JAX programs. Estimates structural live-value memory from JAXPR and compares it with compiler memory accounting. Includes calibrated assessments and batch-size planning.

Experimental estimates are not exact runtime peak-memory predictions or guarantees against out-of-memory errors.

SELECTED WORKSHOP RESEARCH

Authority Firewall studies instruction authority in language-model representations and latent monitoring of prompt-injection and agent-hijacking attempts.

Authority Firewall: Latent Instruction-Hierarchy Monitoring for Safe Personalized Agents

Aleksandar L. Slavov and Cynthia Lambov - equal contribution.

PILA workshop at KDD 2026 | [Workshop paper](#)

Authority Firewall: A Latent-Space Defense Against Prompt Injection and Agent Hijacking

Cynthia Lambov and Aleksandar Slavov.

SafeAI workshop at UAI 2026 | [OpenReview record](#)

AFFILIATION & TECHNICAL FOCUS

Baba Tonka High School of Mathematics, Bulgaria (affiliation listed on the PILA 2026 paper).

Python / PyTorch / JAX / ML debugging / memory analysis / AI safety